B.TECH. PROJECT ON

MACHINE LEARNING FRAMEWORK FOR RF-BASED DETECTION OF UNAUTHORIZED AERIAL SIGNALS

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B.Tech. in
Electronics & Communication Engineering



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CERTIFICATE

This is to certify that Puranjai (2023UEC2510), Harshita (2023UEC2512), Ankit Sharma (2023UEC2539) and Eva Arya (2023UEC2547) have carried out their project work presented in this project entitled "Machine Learning Framework for RF-Based Detection of Unauthorized Aerial Signals" for the award of Bachelor of Technology, Department of Electronics and Communication Engineering, Netaji Subhas University of Technology, New Delhi, under my/our supervision. The project embodies the work carried out by the students during the project period and the contents of the project have been prepared for the partial fulfillment of the B.Tech. degree requirements.

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ABSTRACT

Radio-frequency (RF) fingerprinting is a passive approach for unmanned aerial vehicles (UAVs) detection and identification, based on the features of their wireless transmissions. This work examines the feasibility of reliable drone detection and four-class identification based on a compact engineered RF fingerprint from the DroneRF dataset, and the dependence of the learned classifier to the absolute signal amplitude.

The project is composed of 227 observations at the recording level, obtained by pairing the lower- and upper-band recordings of the DroneRF dataset. The signal processing basis is a fixed python reimplementation of the relevant DroneRF preprocessing pipeline. Sixteen engineered descriptors are extracted from each recording pair covering statistical, waveform, fractal and spectral properties. We then evaluate a 200-tree Random Forest with recording-level stratified 10-fold cross-validation, using training-only preprocessing and feature selection.

The original 16-dimensional fingerprint achieves 97.79% ± 2.21% mean identification accuracy and 100% mean binary detection accuracy under the stated recording-level protocol. Controlled uniform amplitude scaling from 0.25× to 4× reveals substantial degradation in four-class identification away from the nominal scale, while binary detection remains comparatively robust. Three scale-normalized descriptors are therefore constructed analytically, and their empirical performance across the tested scales is reported separately from nominal performance. Finally, leakage-free fold-wise feature selection shows that compact representations can retain performance comparable to the full fingerprint; a representative six-feature set consists of RMS, Variance, Mean Absolute Value, Higuchi Fractal Dimension, Spectral Flatness and Waveform Factor. The report also records the principal limitation of the study: validation is recording-level rather than session-independent.

CONTENT

1 Introduction **9**
1.1 Objectives 9
1.2 Guiding Research Questions 9
1.3 Motivation 10
2 Literature Survey **11**
2.1 RF-Based Drone Detection 11
2.2 Identified Research Gaps 12
3 Methodology **13**
3.1 DroneRF Dataset 13
3.2 Original Preprocessing Reproduction 14
3.3 16-Dimensional RF Fingerprint 15
3.3.1 Statistical Features 15
3.3.2 Waveform and Time-Domain Features 16
3.3.3 Higuchi Fractal Dimension 17
3.3.4 Spectral Features 17
3.4 Classification Framework 18
3.5 Evaluation Protocol and Metrics 18
3.5.1 Memory-Safe Feature Extraction 19
3.5.2 Data Leakage Control 19
4 Results and Scope of Future Work **20**
4.1 Baseline Performance 20
4.2 Amplitude Sensitivity Analysis 20
4.2.1 Bebop versus AR 22
4.3 Scale-Normalized Features 23
4.4 Feature Characterization and Selection 24
4.4.1 Comparison Across Representations 26
4.5 Implementation Summary 27
4.6 Limitations 27
4.7 Future Scope 28
5 Conclusion **29**
6 References **30**

LIST OF FIGURES

1 RF fingerprinting research pipeline used in this project. 10

2 Types of Drones. 13
3 Construction of the recording-level fingerprint dataset from the raw DroneRF files. 14
4 DroneRF classification hierarchy and the scope used in this project. . . . 14
5 Feature categories forming the 16-dimensional RF fingerprint. 15

6 Pooled out-of-fold confusion matrix for four-class identification. 21
7 Four-class identification and binary detection accuracy under uniform amplitude scaling. 22
8 Random Forest feature-importance analysis for the Bebop-versus-AR comparison. 23
9 Feature stability across amplitude scales 24
10 Identification accuracy as a function of the number of selected features. Error bars show the reported fold-wise standard deviation. 25
11 Fingerprint Representation 26

LIST OF TABLES

1 DroneRF recording-level class distribution 13
2 Summary of the 16 engineered RF fingerprint features 18

3 Baseline pooled identification metrics 20
4 Accuracy under controlled amplitude scaling 21
5 Effect of the number of selected features under fold-wise selection . . . 25
6 Representative compact feature set and selection frequency 26
7 Comparison of fingerprint representations under the project protocol . . 27

Chapter 1: Introduction

1.1 Objectives

The main goal of this project is to create and study an RF-based system for finding and recognizing drones. This is done using RF fingerprinting methods on a dataset called DroneRF that's available to the public. The work is about making an RF description and checking how well it works. It also looks closely at the features how sensitive they are to signal strength, how strong they are and how well they can be used in different situations.

The project tries to make an RF fingerprint from recordings in DroneRF by using numbers, time, shapes and frequency parts of the signals that are received. This fingerprint is tested to find drones and figure out what type of drone it is. It also looks more into what kind of drone it is and what flying mode it's in. The features that are taken out are checked to see how good they are at telling things apart, in numbers how they relate to each other if they are not needed and how important each one is.

One big goal is to look at how sensitive the RF fingerprint method is when the signal strength changes. The project also wants to make descriptors that are not affected by how strong the signal is. The project also looks into whether useful and steady features can be picked to make a fingerprint. This should keep the level of identification performance but use less space and less computer power.

Finally, the work evaluates the fingerprint's robustness under varying noise levels, signal to noise ratios, channel conditions, and acquisition setups, as well as its performance across different RF environments. These evaluations aim to guide the development of a compact, robust RF fingerprint, one that uses fewer features, requires less computational power, and generalizes well across diverse operating conditions.

1.2 Guiding Research Questions

The methodology documents articulate five research questions around which the project is organized:

1. **Detection:** Can the presence of a drone be detected from an RF recording? This is a binary classification problem: No Drone versus Drone.

2. **Identification:** Can the specific drone platform be identified? This is a four-class task: No Drone, Bebop, AR and Phantom.

3. **Attribution:** Which signal characteristics are associated with correct classification? This is studied using amplitude/statistical descriptors, waveform descriptors, spectral

descriptors, fractal complexity and inter-feature relationships.

4. **Amplitude sensitivity:** How much does identification performance change when amplitude-related quantities are uniformly scaled by 0.25×, 0.5×, 0.75×, 1×, 1.5×, 2× and 4×?

5. **Compactness and robustness:** Can mathematically scale-normalized descriptors and training-fold-only feature selection produce a smaller representation without losing the useful properties of the full fingerprint?

1.3 Motivation

RF-based drone analysis is appealing because the sensing can be done passively and because the signal holds data that cannot be seen with cameras. Still, a received RF signal is influenced by how it is captured. The absolute amplitude can shift with gain, propagation loss, antenna setup, orientation, and other aspects. Therefore a model that works well on one set of recordings may rely on amplitude patterns that do not stay the same when the scale changes.

Another reason is computation. The raw DroneRF recordings are big. The fingerprint in this project has only sixteen numbers for each recording. A small representation can make storage, transmission, and classifier input easier, although shrinking the fingerprint size does not mean the total signal-processing effort is smaller. For example, HFD and spectral descriptors still need to be computed from the recording.

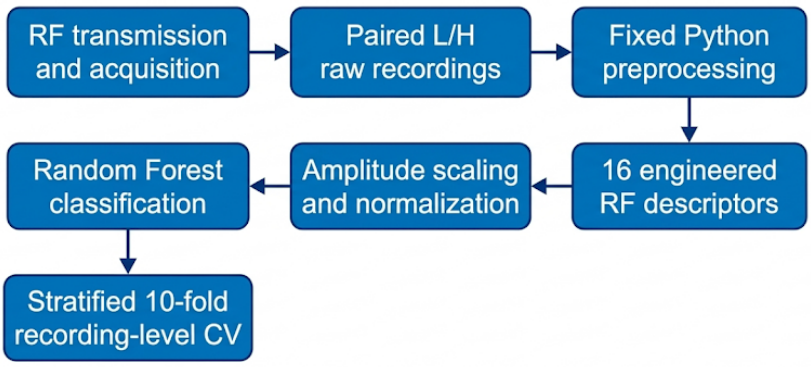


Fig. 1: RF fingerprinting research pipeline used in this project.

Chapter 2: Literature Survey

2.1 RF-Based Drone Detection

Existing RF-based drone detection studies show that classical machine-learning methods can work effectively when combined with engineered RF descriptors. Zhang et al. (2021) and Medaiyese et al. (2021) explored machine-learning approaches for RF-based drone detection and identification. The literature reviewed for this project also includes approaches based on Random Forest, SVM and KNN, time-frequency representations with CNNs, hierarchical feature learning with XGBoost, and WiFi/RF signatures for distinguishing UAV platforms. Xiao and Zhang (2021) investigated RF signatures for micro-UAV detection and identification, while Nie et al. (2021) examined UAV detection using WiFi signals and RF fingerprints. These studies show that RF information can support both drone detection and identification, although they differ in representation size, computational requirements and evaluation conditions.

Classical machine-learning approaches are particularly relevant to the present work because the final RF fingerprint contains only sixteen engineered values per recording. Random Forest is well suited to this type of tabular data because it can model nonlinear relationships, capture feature interactions and provide feature-importance measures. It can also be trained on a relatively small dataset without requiring a deep neural-network architecture.

Mandal and Satija (2023) demonstrated the use of a time-frequency multiscale CNN for RF-based drone detection and identification, highlighting the benefit of combining complementary RF representations. However, such approaches generally involve more computationally demanding training and inference pipelines than the compact tabular approach used in this project. Other studies also investigate the effect of noise and SNR on RF-based detection, which motivates the future extension of the present work toward controlled noise and channel experiments.

The DroneRF dataset was introduced by Al-Sa'd et al. (2019) as an open RF drone database containing three drone platforms along with background RF activity. The current project uses the first two classification levels from this dataset: binary detection and four-class identification. The ten-class flight-mode classification task remains outside the scope of the current experiments and is therefore considered future work.

2.2 Identified Research Gaps

The uploaded methodology documents highlight the primary research gap as the connection between strong nominal fingerprint performance and dependence on absolute amplitude, the project therefore studies amplitude sensitivity explicitly instead of positing high clean-data accuracy as sufficient evidence of robustness.

A second gap concerns compactness, instead of removing features solely because they are correlated, the project performs feature selection inside each training fold and then evaluates the selected representation on the held-out fold. This separates the feature selection decision from the test data and avoids the direct leakage that would arise picking a feature set using the entire set first and then using cross-validation.

The resulting research contributions should be carefully described. The individual descriptors used are established signal processing features, the methodological contribution of the project is the combined experiment framework: fixed preprocessing, engineered recording level fingerprinting, amplitude sensitivity analysis, scale-normalized descriptors, redundancy analysis and leakage-controlled fold-wise feature selection.

Chapter 3: Methodology

3.1 DroneRF Dataset



Fig. 2: Types of Drones.

The dataset contains three drone platforms along with background RF activity: Parrot Bebop, Parrot AR Drone and DJI Phantom 3. The recording-level class distribution used in this project is shown below.

Table 1: DroneRF recording-level class distribution

Class	**Label**	**Recordings**
No Drone / Background	0	41
Bebop	1	84
AR	2	81
Phantom	3	21
Total		**227**

The raw database has 454 CSV files, representing 227 recording-level L/H pairs. Each pair comes from the same physical acquisition, so the two files are not independent observations:

$$227 \text{ recording pairs} \times 2 \text{ files per pair} = 454 \text{ raw files}. \tag{1}$$

The original preprocessing uses a segmentation factor of 100, producing $22,700$ derived segments. These are not $22,700$ independent recordings because several

segments can come from the same acquisition. For this project, one L/H pair is treated as one observation, resulting in 227 recording-level fingerprints.

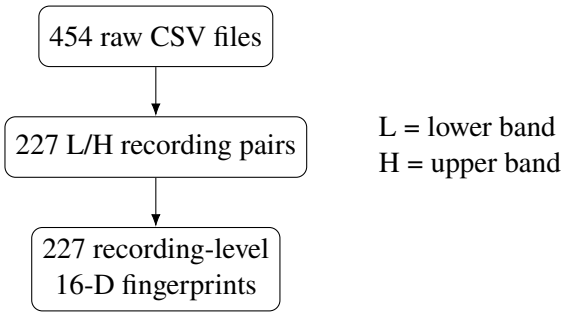


Fig. 3: Construction of the recording-level fingerprint dataset from the raw DroneRF files.

The BUI mapping is Background = 00000; Bebop = 10000, 10001, 10010, 10011; AR = 10100, 10101, 10110, 10111; and Phantom = 11000. Files are indexed recursively and paired using both BUI and recording index, ensuring that an H file from one recording is not combined with an L file from another.

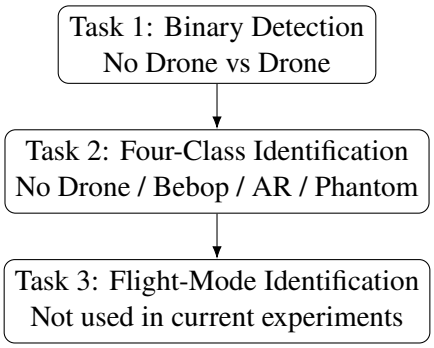


Fig. 4: DroneRF classification hierarchy and the scope used in this project.

3.2 Original Preprocessing Reproduction

The original DroneRF repository describes three main stages [7] : aggregation and resegmentation, labeling and normalization, and classification. In this project, the relevant signal-processing stages were reproduced in Python instead of directly using the original MATLAB implementation.

The preprocessing sequence is: recursively index raw L/H CSV files, pair recordings, select and segment the data, subtract the mean, compute the FFT, retain the positive/non-redundant spectrum, estimate the relative L/H scale, stitch the two spectral portions, and apply the established normalization. The FFT representation length is fixed at $M = 2048$, following the original pipeline.

For L/H stitching, $Q = 10$ boundary bins are used to estimate the relative scale:

$$c = \frac{\text{mean}(L_{\text{last } Q \text{ bins}})}{\text{mean}(H_{\text{first } Q \text{ bins}})}. \tag{2}$$

The H spectrum is then multiplied by c before concatenation, reducing the artificial discontinuity caused by different amplitude scales in the two separately captured portions.

The preprocessing was kept fixed for all subsequent experiments so that changes in classification performance could be attributed to the engineered fingerprint representation rather than simultaneous changes in signal preprocessing. The proposed 16-dimensional fingerprint was subsequently extracted from the paired L/H recordings using statistical, time-domain, fractal, and spectral descriptors, providing a compact representation for classification.

3.3 16-Dimensional RF Fingerprint

For a recording with samples $x_1, x_2, \ldots, x_N$, the engineered representation converts the raw sequence into sixteen descriptive statistics grouped into statistical, waveform/time-domain, fractal/complexity and spectral categories.

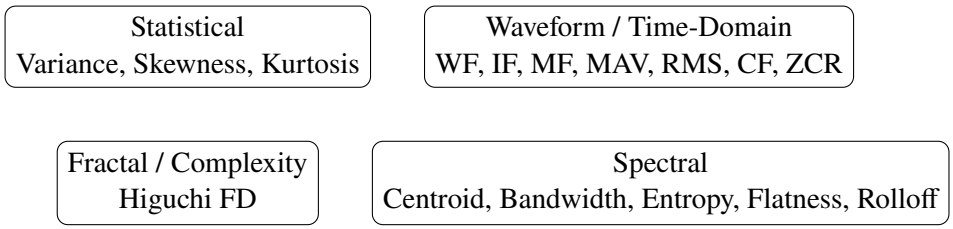


Fig. 5: Feature categories forming the 16-dimensional RF fingerprint.

3.3.1 Statistical Features

Variance measures dispersion around the mean:

$$\text{Var}(x) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2, \qquad \mu = \frac{1}{N}\sum_{i=1}^{N} x_i. \tag{3}$$

It is amplitude-sensitive, with $\text{Var}(ax) = a^2\,\text{Var}(x)$.

Skewness measures distribution asymmetry:

$$\text{Skewness} = \frac{E[(X-\mu)^3]}{\sigma^3}. \tag{4}$$

Its class separation was weak ($p = 0.571$, $\eta^2 \approx 0.0059$), and it is invariant to uniform

scaling.

The Pearson kurtosis used here is

$$K = \frac{E[(X - \mu)^4]}{\sigma^4}. \tag{5}$$

Kurtosis is scale-invariant and showed meaningful class separation with $\eta^2 \approx 0.345$.

3.3.2 Waveform and Time-Domain Features

The waveform factor is

$$WF = \frac{RMS}{MAV}, \tag{6}$$

where

$$RMS = \sqrt{\frac{1}{N}\sum_{i=1}^{N} x_i^2}, \qquad MAV = \frac{1}{N}\sum_{i=1}^{N} |x_i|. \tag{7}$$

Both quantities scale linearly with $|a|$, so their ratio is scale-invariant. The observed waveform factor showed substantial class separation ($\eta^2 \approx 0.548$).

The impulse factor and margin factor are

$$IF = \frac{\max_i |x_i|}{MAV}, \tag{8}$$

and

$$MF = \frac{\max_i |x_i|}{\left(\text{mean}(\sqrt{|x_i|})\right)^2}. \tag{9}$$

Both are scale-invariant; their documented effect sizes were approximately $\eta^2 \approx 0.326$ and $\eta^2 \approx 0.289$, respectively.

MAV and RMS provide direct measures of signal magnitude. RMS showed particularly strong class separation ($\eta^2 \approx 0.792$). Variance and RMS were strongly correlated (Pearson ≈ 0.937, Spearman ≈ 0.999), although they describe different properties: variance measures dispersion around the mean, while RMS measures overall quadratic magnitude.

The crest factor is

$$CF = \frac{\max_i |x_i|}{RMS}, \tag{10}$$

with $\eta^2 \approx 0.341$. Zero-crossing rate is approximated by

$$ZCR \approx \frac{\text{number of sign changes}}{N - 1}. \tag{11}$$

For positive uniform scaling it remains unchanged, and its effect size was comparatively modest ($\eta^2 \approx 0.137$).

3.3.3 Higuchi Fractal Dimension

Higuchi Fractal Dimension (HFD) captures waveform complexity or roughness rather than absolute magnitude. It follows the relationship $L(k) \propto k^{-D}$, where D is related to fractal dimension, and is approximately invariant to uniform vertical scaling.

HFD showed the strongest documented class separation ($\eta^2 \approx 0.814$) and was selected in all ten training folds during feature selection. To keep computation practical, it was calculated on a uniformly sampled subset rather than the complete recording, using HFD_POINTS = 25,000 and KMAX = 30.

3.3.4 Spectral Features

Spectral centroid is defined as

$$SC = \frac{\sum_k f_k P_k}{\sum_k P_k}, \tag{12}$$

where P_k is spectral magnitude/energy. It represents the centre of the spectral distribution rather than a literal carrier frequency, with $\eta^2 \approx 0.198$.

Spectral bandwidth measures the spread around the centroid:

$$SB = \sqrt{\frac{\sum_k P_k (f_k - SC)^2}{\sum_k P_k}}, \tag{13}$$

with $\eta^2 \approx 0.187$.

For normalized spectral probabilities $p_k = P_k / \sum_i P_i$, spectral entropy is

$$H = -\sum_k p_k \log p_k, \qquad H_{norm} = \frac{H}{\log N}. \tag{14}$$

It showed substantial class-dependent structure, with $\eta^2 \approx 0.489$.

Spectral flatness is

$$SF = \frac{(\prod_k P_k)^{1/N}}{\frac{1}{N} \sum_k P_k}, \tag{15}$$

with $\eta^2 \approx 0.477$. It appeared in 9 of 10 selected feature sets. The approximate class means were 0.918 for background, 0.911 for Bebop, 0.887 for AR and 0.965 for Phantom.

Spectral rolloff is the frequency at which a selected cumulative-energy threshold is reached. Since common scaling cancels in the normalized cumulative distribution, rolloff is scale-invariant; its documented effect size was $\eta^2 \approx 0.231$.

Table 2: Summary of the 16 engineered RF fingerprint features

Category	**Features**	**Count**
Statistical	Variance, Skewness, Kurtosis	3
Waveform / time-domain	WF, IF, MF, MAV, RMS, CF, ZCR	7
Fractal / complexity	HFD	1
Spectral	Centroid, Bandwidth, Entropy, Flatness, Rolloff	5
Total		**16**

3.4 Classification Framework

The main classifier is a Random Forest with 200 trees:

```
RandomForestClassifier(n_estimators=200, random_state=42, n_jobs=-1).
```
(16)

The trees are built using bootstrap samples of the data and a subset of features selected at random. The results of all the trees are then combined, using majority voting, to arrive at a final prediction. The random state was fixed for the sake of reproducibility. This works as long as the same data and software are used.

The random forest model does not require the features to be scaled, since decision trees are not sensitive to the scale of the features. Nevertheless, StandardScaler was added to the pipeline, since SVM and KNN, which were also evaluated for the task, do require scaling. The scaler is fit only on the training part of each fold.

3.5 Evaluation Protocol and Metrics

Evaluation uses recording-level stratified 10-fold cross-validation. Each recording appears once in a held-out fold, while stratification approximately preserves class proportions. This is useful with only 227 observations and the class counts of 41, 84, 81 and 21.

Accuracy is defined as

$$Accuracy = \frac{TP + TN}{\text{Total Samples}}. \tag{17}$$

For multi-class classification, precision and recall are computed per class, and

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}. \tag{18}$$

Macro-F1 gives equal weight to each class, whereas weighted F1 accounts for class support. Because the dataset is imbalanced, accuracy is considered together with class-wise and F1-based measures.

The feature matrix and labels are

$$X \in \mathbb{R}^{227\times 16}, \qquad y \in \{0,1,2,3\}^{227} \tag{19}$$

for four-class identification, and $y \in \{0,1\}^{227}$ for binary detection.

3.5.1 Memory-Safe Feature Extraction

A raw recording can contain roughly ten million samples, so each CSV is processed incrementally rather than loaded completely into memory. Running sums, sums of squares and counts are used for mean and variance, while skewness and kurtosis are also computed using streaming statistics.

ZCR uses an additional streaming pass so that sign changes across chunk boundaries are not missed. HFD uses a uniformly sampled 25,000-point subset with $K_{max} = 30$. Spectral descriptors use 250,000 uniformly sampled points and an FFT size of 8192. These are computational approximations for descriptor extraction and should not be interpreted as an exact reconstruction of the full physical spectrum.

For each L/H pair, the lower- and upper-band files are independently converted into feature vectors and then averaged feature-by-feature:

$$F_{final} = \frac{F_L + F_H}{2}. \tag{20}$$

Thus, each row of the fingerprint CSV represents one recording, one L/H pair and one sixteen-dimensional fingerprint.

3.5.2 Data Leakage Control

To prevent the information from the held-out fold from leaking into the training process, the scaler and feature-selection decisions are learned only from the training part of each fold. Using the entire dataset to calculate normalization statistics or feature importance before cross-validation would leak information from the test data.

Within each fold, we train a Random Forest on the training data, calculate feature importances, rank features, select the top-k, and evaluate them on the held-out fold. It is this training-only selection, within an outer stratified 10-fold evaluation, that is not nested cross-validation.

Chapter 4: Results and Scope of Future Work

4.1 Baseline Performance

Using the full sixteen-dimensional fingerprint and the 200-tree Random Forest, recording-level stratified 10-fold cross-validation yielded a mean four-class identification accuracy of 97.79 % (standard deviation = 2.21 percentage points) and a macro-F1 of approximately 0.985.

The pooled out-of-fold confusion matrix has 222 correct predictions out of 227, resulting in approximately 97.80% pooled accuracy. The two accuracies reported are not contradictory: 97.79 % is the average of the ten fold accuracies, whereas 97.80 % is obtained by concatenating all held-out predictions and computing accuracy once.

The pooled matrix shows 100% recall for No Drone and Phantom classes. Bebop has $82/84 \approx 97.62\%$ Recall. Two of Bebop's errors are attributed to misclassification as AR. AR has recall $78/81 \approx 96.30\%$, with three errors misclassified as Bebop. The same representation and classifier achieved 100% mean accuracy with 0% standard deviation over the ten folds for the simpler binary detection task. This establishes perfect detection on the stated DroneRF recording-level cross-validation protocol but does not establish perfect detection under arbitrary conditions.

Table 3: Baseline pooled identification metrics

Class	Precision	Recall	F1	Support
No Drone	1.000	1.000	1.000	41
Bebop	0.976	0.976	0.976	84
AR	0.975	0.963	0.969	81
Phantom	1.000	1.000	1.000	21

4.2 Amplitude Sensitivity Analysis

Only three of the sixteen descriptors are directly amplitude sensitive under uniform scaling: Variance, Mean Absolute Value and RMS. MAV and RMS scale linearly, Variance scales quadratically Many of the remaining descriptors are ratio or distribution shape descriptors and are theoretically invariant, or approximately invariant in the case of HFD, under uniform scaling.

The motivation, for testing amplitude sensitivity is straightforward. A nominal accuracy alone does not prove that the classifier learns only intrinsic transmitter characteristics. The controlled experiment applies amplitude sensitivity testing

$$x'[n] = a\,x[n] \tag{21}$$

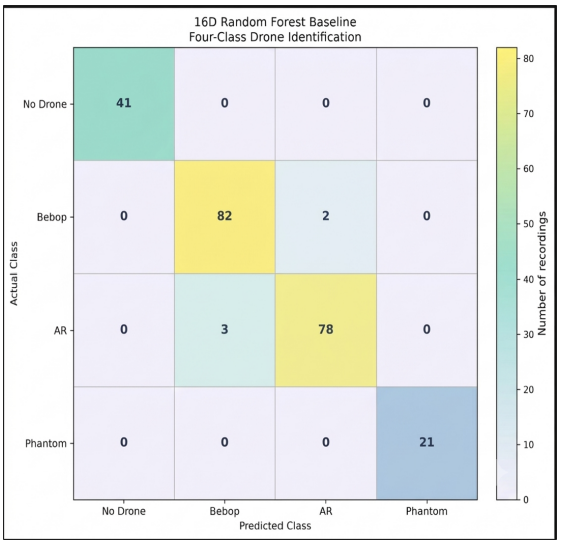


Fig. 6: Pooled out-of-fold confusion matrix for four-class identification.

to the amplitude-related quantities using $a \in \{0.25, 0.5, 0.75, 1, 1.5, 2, 4\}$. This is a synthetic uniform scaling experiment; it does not simulate distance, multipath, fading, receiver gain nonlinearities, Doppler, additive noise or antenna variation.

Table 4: Accuracy under controlled amplitude scaling

Scale	Identification Accuracy (%)	Detection Accuracy (%)
0.25×	47.61	81.01
0.50×	77.55	97.79
0.75×	95.16	99.55
1.00×	97.79	100.00
1.50×	86.28	100.00
2.00×	80.14	99.57
4.00×	62.94	91.17

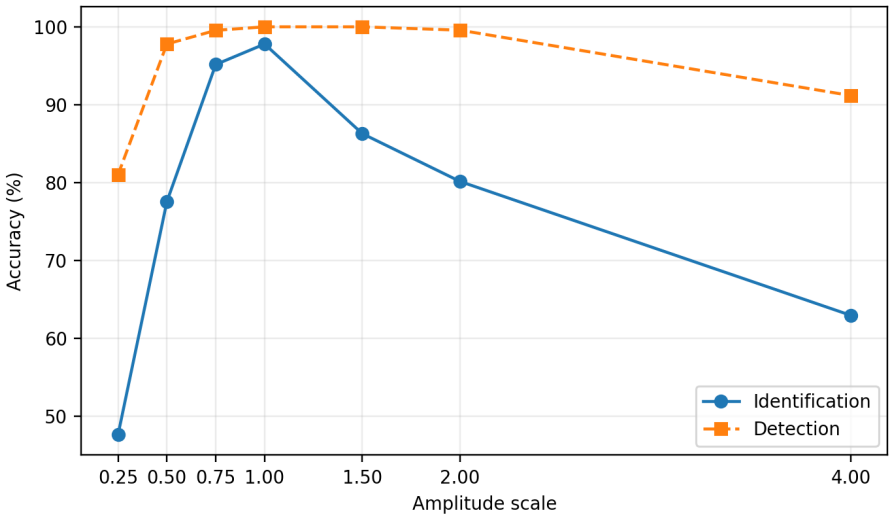


Fig. 7: Four-class identification and binary detection accuracy under uniform amplitude scaling.

Identification accuracy is highest when the scale is close to 1. As the scale moves away from 1 accuracy starts to drop. The difference between 0.25× and 4× is not due, to some rule that has not been tested. It comes from how the features are distributed after training. When scaling changes the size of inputs, the feature vectors end up in parts of the feature space that do not match the decision boundaries the model learned during training. This causes performance to degrade.

4.2.1 Bebop versus AR

Because all baseline identification errors occur between Bebop and AR, a focused pairwise experiment was conducted on their 165 combined recordings. Pairwise accuracy was 96.95% at 1×, 49.12% at 0.25×, and 66.03% at 4×, with intermediate values of 73.86% at 0.5×, 93.86% at 0.75×, 92.72% at 1.5× and 86.03% at 2×.

The feature importance analysis using Random Forest for this comparison task resulted in RMS = 0.288, Variance = 0.283 and MeanAbs = 0.191. These three descriptors that are sensitive to amplitude make up, about 76.2% of the importance in this analysis. This shows that information related to amplitude plays a role in the separation of Bebop and AR. The feature importance should not be seen as a measure of how much each feature contributes to accuracy.

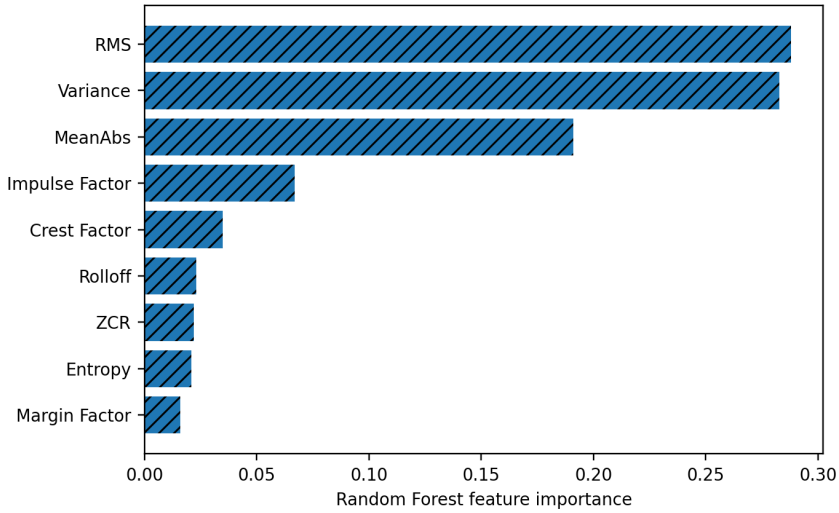


Fig. 8: Random Forest feature-importance analysis for the Bebop-versus-AR comparison.

4.3 Scale-Normalized Features

The goal of scale normalization is not to say that amplitude has no meaning. The actual size of the amplitude is definitely important when things are measured the way they are supposed to be. The real question is whether we can keep the information, about different classes while making the results less affected by the exact size of the numbers.

Three normalized descriptors are constructed:

$$V_{norm} = \frac{Variance}{RMS^2}, \tag{22}$$

$$MA_{norm} = \frac{MeanAbs}{RMS}, \tag{23}$$

and

$$R_{MA} = \frac{RMS}{MeanAbs}. \tag{24}$$

For example, because $RMS(ax) = |a|RMS(x)$ and $MeanAbs(ax) = |a|MeanAbs(x)$,

$$\frac{RMS(ax)}{MeanAbs(ax)} = \frac{|a|RMS(x)}{|a|MeanAbs(x)} = \frac{RMS(x)}{MeanAbs(x)}. \tag{25}$$

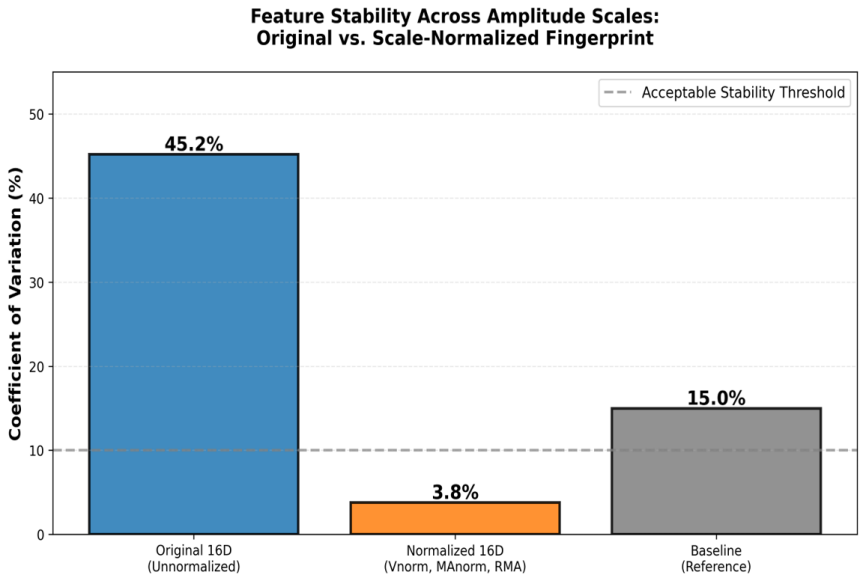


Fig. 9: Feature stability across amplitude scales

The cancellation is an algebraic property of the construction. The empirical question is whether the resulting representation is sufficiently discriminative on held-out recordings.

Across the tested amplitude scales, the scale-normalized representation achieved 96.46% ± 4.06% identification accuracy. In the nominal comparison table, the scale-normalized 16-D representation achieved 96.90% ± 3.66% under the same recording-level evaluation protocol. These numbers answer different aggregation questions and must not be conflated: the first summarizes performance across the controlled amplitude-scaling experiment, while the second is the nominal cross-validation comparison.

The normalized representation therefore trades some nominal discrimination for greater stability under the specific synthetic amplitude perturbation. The result supports the narrower conclusion that mathematically scale-normalized descriptors can reduce sensitivity to uniform amplitude scaling; it does not establish robustness to arbitrary channel or receiver changes.

4.4 Feature Characterization and Selection

Feature selection was done because the sixteen-dimensional fingerprint might have redundant or overlapping information. If two variables are highly correlated, that's not a good enough reason to throw one of them away. RMS, Variance and MeanAbs are very closely related but are different mathematical quantities, and tree-based models can take advantage of the interactions between correlated descriptors.

For each fold, feature selection was restricted to the training data. A Random Forest

was trained, feature importances were extracted and the top-k descriptors were selected. The values of k examined were 4, 6, 8, 10, 12 and 16.

Table 5: Effect of the number of selected features under fold-wise selection

Features selected	Identification accuracy (%)
4	97.79 ± 2.33
6	97.79 ± 2.33
8	97.79 ± 2.33
10	97.79 ± 2.33
12	98.22 ± 2.30
16	97.79 ± 2.33

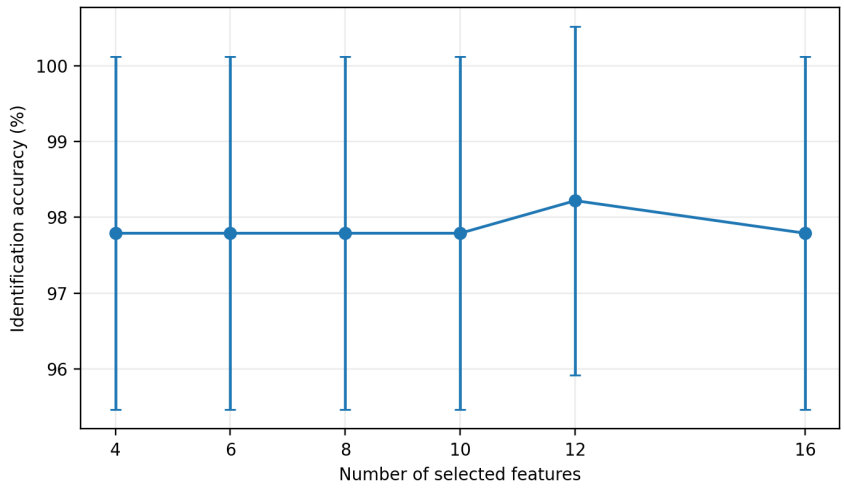


Fig. 10: Identification accuracy as a function of the number of selected features. Error bars show the reported fold-wise standard deviation.

The six-feature selection procedure gives 97.79% ± 2.33%, essentially matching the full representation under this fold-wise protocol. The dimensionality reduction is

$$\frac{16-6}{16} \times 100\% = 62.5\%. \tag{26}$$

This is a reduction in fingerprint dimensionality, not a claim of a 62.5% reduction in total computational cost.

Selection stability was also measured. RMS, Variance, MeanAbs and HFD were selected in all 10 folds, while Spectral Flatness and Waveform Factor were selected in 9 of 10 folds. The representative compact set is therefore:

- Root Mean Square (RMS),
- Variance,
- Mean Absolute Value (MeanAbs),

- Higuchi Fractal Dimension (HFD),
- Spectral Flatness, and
- Waveform Factor.

This set encompasses absolute amplitude, waveform shape, temporal complexity and spectral structure. A six-feature fixed set chosen after examining all folds was assessed at 97.35% ± 2.28%, however this number is subject to post-selection bias and should not be provided as an independently validated estimate. The correct methodologically compact result is the fold-wise 97.79% ± 2.33% figure.

Table 6: Representative compact feature set and selection frequency

Feature	Selection Frequency
RMS	10/10
Variance	10/10
MeanAbs	10/10
Higuchi Fractal Dimension	10/10
Spectral Flatness	9/10
Waveform Factor	9/10

4.4.1 Comparison Across Representations

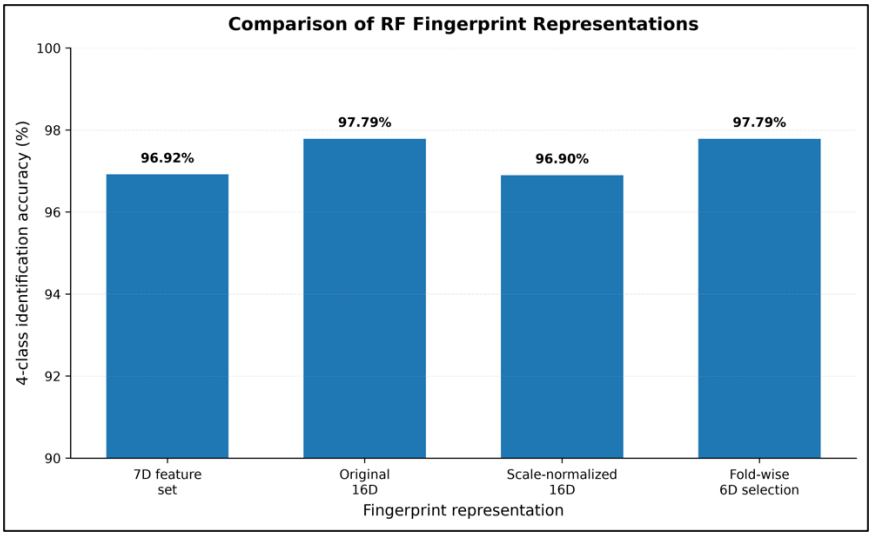


Fig. 11: Fingerprint Representation

An external seven-dimensional feature set from a 2024 fingerprinting paper [1] was evaluated on the present DroneRF fingerprint dataset using the same Random Forest protocol. It achieved 96.92% ± 3.60%. This is an evaluation performed by the present project and is not the original paper's reported result.

Table 7: Comparison of fingerprint representations under the project protocol

Representation	Identification Accuracy
External 7-D feature set evaluated on this dataset/protocol	96.92 ± 3.60%
Original 16-D fingerprint	97.79 ± 2.21%
Scale-normalized 16-D fingerprint	96.90 ± 3.66%
Fold-wise compact selection (6-D)	97.79 ± 2.33%

4.5 Implementation Summary

The whole experimental pipeline may be summarized as follows. First, the raw DroneRF directory is indexed recursively, with the L/H files matched by BUI and recording index, and the established preprocessing is re-implemented in Python. The preprocessing is then frozen with respect to the original DroneRF pipeline [7] as of the Python reproduction stage. For each L/H pair, the 16-dimensional fingerprint is computed as the concatenation of the L and H vectors, averaged across the two, and then the 227-by-16 matrix is classified using the 200-tree Random Forest, with standardization learned within each fold. The predictions are finally aggregated over the left out fold, with ten such metrics combined across the folds. The same pipeline is then applied to the amplitude-scaled data, with the scale-normalized descriptors and fold-specific feature selection employed.

4.6 Limitations

The main limitation is the unit of validation. The data set does not provide the explicit session identifiers that could be used for the grouped, session-independent cross-validation. Thus, the reported results are recording-level, stratified 10-fold cross-validation results. Recordings that originated from the same acquisition session could possibly appear in different folds. Thus, these results cannot be interpreted as the evidence for session-independent generalization.

The second limitation is the amplitude experiment. Uniform multiplication is a controlled synthetic transformation, which does not capture all effects that could alter the received RF amplitude and waveform structure. The experiment was designed to isolate the dependence on the overall magnitude, rather than capturing the real propagation channel.

The study does not establish any universal drone identification, receiver independent, robustness to arbitrary SNR, robustness to arbitrary channels, and deployment-ready-to-use performance. The six-feature set is empirically useful on this data set and protocol, but it is not claimed to be universally optimal. Finally, reducing the fingerprint dimensionality by 62.5% does not necessarily imply an equivalent reduction

in the total runtime because raw signal processing, HFD and spectral extraction remain part of the pipeline.

4.7 Future Scope

Several extensions follow directly from the limitations and research questions.

1. **Noise and SNR robustness:** Evaluate the 16-D original, scale-normalized and compact representations under controlled SNR conditions, for example from 0 dB to 20 dB, and identify features that remain stable.
2. **Flight-mode identification:** Extend the fingerprint to the ten-class flight-mode task available in the original DroneRF framework and determine whether the compact representation remains sufficient.
3. **Channel and acquisition robustness:** Test different antenna configurations, receiver gains and propagation environments, while explicitly separating channel effects from transmitter characteristics.
4. **Cross-condition generalization:** Train under one acquisition condition and test under another to quantify distribution shift.
5. **Open-set recognition:** Investigate how the system should respond when an unseen drone model is encountered rather than forcing it into one of the known classes.

Chapter 5: Conclusion

This report presents a work which engineered and assessed a compact engineered RF fingerprint for drone detection and identification using the DroneRF dataset. The signal-processing baseline was reimplemented in Python and frozen for downstream experiments, isolating the fingerprint design effect from changes in preprocessing.

The full sixteen-dimensional representation is built from statistical, waveform, fractal and spectral descriptors. Under recording-level stratified 10-fold cross-validation, the 200-tree Random Forest achieved 97.79% ± 2.21% four-class identification accuracy and 100% mean binary detection accuracy. The pooled confusion matrix then indicated that the principle identification challenge was distinguishing Bebop from AR.

The amplitude-sensitivity experiment provided an important qualification to the nominal baseline. Identification accuracy dropped from 97.79% at the original scale to 47.61% at 0.25× and 62.94% at 4×, whereas binary detection remained comparatively robust. Furthermore, the feature-importance analysis of the Bebop-versus-AR problem indicated that RMS, Variance and MeanAbs together accounted for approximately 76.2% of the impurity-based feature importance. The above findings thus support the conclusion that the classifier makes substantial use of absolute amplitude information under the studied acquisition conditions.

Three scale-normalized descriptors were then constructed so that the uniform amplitude factor cancels algebraically. Across the controlled amplitude-scaling experiment, the normalized representation achieved 96.46% ± 4.06%, illustrating a trade-off between nominal discrimination and robustness to the particular amplitude perturbation tested.

Finally, fold-wise feature selection indicated that a six-feature representation can retain essentially the same identification performance as the full fingerprint under the stated protocol. A representative set of RMS, Variance, MeanAbs, HFD, Spectral Flatness and Waveform Factor, thus reduces the fingerprint dimensionality by 62.5% while spanning multiple types of signal information. This result should be understood as an empirical finding for the DroneRF dataset and the present evaluation protocol, not a universal statement about the minimum number of features required for drone identification.

Overall, the project demonstrates a coherent methodology for moving from raw RF recordings to an interpretable compact fingerprint, while explicitly examining amplitude dependence, feature redundancy and validation leakage. The principal next step is stronger generalization testing across sessions, channels, SNR conditions and unseen drone types.

Chapter 6: References

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